Total Control — TwinSAFE Software Change Requirements

Dynamometer System — Operating Modes & Safety Logic

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1. Context

Following the integration session and subsequent review with the Mentee software team, the mode structure and terminology for the Dynamometer System safety controller have been updated. The changes below are required to the TwinSAFE safety software and must be implemented by Total Control.

This document is issued for review. Total Control is requested to review the requirements, raise any concerns, and provide an implementation plan and estimate.

System architecture: the system comprises two units — the Dynamometer Bench (aluminium frame, transparent cover, motors, sensors, control panel) and the Power Rack (Keysight power supplies, Beckhoff drives, TwinSAFE, relay modules).

2. Terminology

The following terms are adopted across all documentation and software going forward:

Old Term	New Term	Notes
Standard Safe Mode	**Run Mode**	Renamed
Standard R&D Mode	**R&D Mode**	Renamed
(new)	**Safe Mode**	New default de-energised state
Emergency Mode	Emergency Mode	Unchanged

3. Control Panel

The Dynamometer Bench has a single control panel on the left side with three buttons:

- EMERGENCY STOP — Red mushroom button with red indicator LED.
- SAFETY RESET — Green button. Functionally the down-step / energy-management control.
- RELEASE LOCK — Blue button. Opens the cover; also forms the dual-press activation gesture when held together with SAFETY RESET.

Gesture rules (applied consistently across all transitions):

- Dual press (SAFETY RESET + RELEASE LOCK together) = step UP one level / activate (add energy).
- Short press SAFETY RESET = step DOWN one level (shed energy).
- RELEASE LOCK alone = open cover, in any state where opening is permitted. Always required to open the cover.
- EMERGENCY STOP = emergency.

Note: stepping UP (adding energy) may be requested physically or by the PLC, but is executed only when the cover interlock reports closed + locked. Opening the cover is physical-only. See the governing principle in Section 5.

4. Operating States

The system has four operating states plus the latched Emergency state. Cover opening is governed by the state; the RELEASE LOCK (blue) button is always the mechanism by which the cover is opened where permitted.

State	Energy	Cover openable?	How reached
Emergency	De-energised, latched	No	Physical E-Stop, PLC request, Mentee driver request, or watchdog loss
Safe Mode	De-energised	Yes (via RELEASE LOCK)	Dis-engage the physical E-Stop (from Emergency); or step-down from R&D
R&D Mode	LV domain on (PS1 + battery + BMS); HV off	Yes (via RELEASE LOCK)	Dual press (SAFETY RESET + RELEASE LOCK) or PLC request from Safe Mode — cover must be closed + locked
Run Mode	Full power (LV + HV / PS2)	No (locked)	Dual press or PLC request from R&D Mode — only if cover closed + locked

State detail

- Safe Mode (default de-energised): PS1, PS2, battery and BMS all de-energised. Cover may be opened via RELEASE LOCK.
- R&D Mode: PS1 (Keysight RP5913A, 48V) and the full LV domain (battery + BMS relays) energised; PS2 (Keysight RP5926A, 800V) disabled and its HEALTH input ignored. Cover may be opened via RELEASE LOCK.
- Run Mode: full power including PS2 (800V). Cover closed and locked; RELEASE LOCK rejected.
- In standard operation the system may move freely between states (subject to the interlocks in Section 5) without activating Emergency Mode.

5. State Transitions

Governing safety principle

The PLC may request any state transition; the TwinSAFE controller decides whether to execute it, and the PLC can never bypass these safety gates. TwinSAFE filters every request against its own conditions, identically to the physical buttons: UP-steps (adding energy) are executed only when the cover interlock reports CLOSED and LOCKED, and are rejected — ignored — if the cover is open; leaving Run Mode requires confirmed zero motor speed and zero current. Opening the cover and clearing an emergency remain physical-only and are never exposed to the PLC.

Applying this principle: UP-steps (Safe → R&D, R&D → Run) may be requested physically or by the PLC, but execute only with the cover closed + locked. DOWN-steps (Run → R&D, R&D → Safe) and Emergency may be requested physically or by the PLC. Opening the cover and clearing an emergency are physical-only. TwinSAFE also reports cover/latch status to the PLC so the requesting software can check the gate before issuing an UP request.

Transition	Gesture	Via PLC	Precondition / Interlock
Emergency → Safe	Engage / dis-engage physical E-Stop	**No — physical only**	Clears the emergency latch and returns to the de-energised Safe Mode. Performed identically whether or not the mushroom was pressed. System remains de-energised; cover stays shut until RELEASE LOCK is pressed.
Safe → R&D (UP)	Dual press (SAFETY RESET + RELEASE LOCK) or PLC request	**Yes**	Adds energy (energises full LV domain: PS1 + battery + BMS). Executed by TwinSAFE only if cover CLOSED + LOCKED; rejected if cover open — same for button or PLC. HV remains off.
R&D → Run (UP)	Dual press (SAFETY RESET + RELEASE LOCK) or PLC request	**Yes — cover-gated**	Adds energy (energises HV / PS2). Executed by TwinSAFE only if cover CLOSED + LOCKED; rejected if cover open — same for button or PLC.
Run → R&D (DOWN)	Short press SAFETY RESET	**Yes**	Sheds energy (drops HV). System must confirm ZERO motor speed and ZERO current before HV is dropped. Applies equally to the PLC path.
R&D → Safe (DOWN)	Short press SAFETY RESET	**Yes**	Sheds energy (de-energises LV domain). Returns to de-energised Safe Mode.
Open cover	RELEASE LOCK alone	**No — physical only**	Exposes the operator. Honoured only where cover opening is permitted (Safe, R&D). Always required to open the cover. Rejected in Run and Emergency.

Any → Emergency	E-Stop / PLC / driver / watchdog	**Yes**	Sheds all energy. Full safety protocol. Latching — recovery is physical (Section 6).

6. Emergency Entry & Recovery

Emergency entry

Emergency Mode may be entered by any of the following. In all cases the result is identical and latching:

- Physical EMERGENCY STOP button press (hardwired safety function — unchanged).
- PLC request (CMD_TRIGGER_EMERGENCY), executed by TwinSAFE.
- Mentee driver request, executed by TwinSAFE.
- Loss of the PLC–TwinSAFE watchdog / heartbeat (Section 7).

When emergency is triggered by a means other than the physical button, the E-Stop mushroom will NOT be in the latched/pressed position. Recovery nonetheless requires deliberate physical action at the machine, as below.

Recovery sequence

The energising and access steps of recovery are physical only and cannot be performed remotely, preventing remote re-energising of a system someone may be working inside. The sequence is:

1. Engage and dis-engage the physical EMERGENCY STOP button. This clears the emergency latch and returns the system directly to the de-energised Safe Mode. Performed identically whether or not the mushroom was already pressed. Physical only.
2. Open the cover with RELEASE LOCK if access is required (physical only). The system remains de-energised.
3. To resume operation, dual press SAFETY RESET + RELEASE LOCK to enter R&D Mode (LV domain up), then dual press again to enter Run Mode (HV up; requires cover closed + locked). Both UP-steps are physical only.

The cover remains locked throughout Emergency, and becomes openable only on reaching Safe Mode — and then only via the RELEASE LOCK button.

7. PLC – TwinSAFE Watchdog / Heartbeat

A continuous watchdog / heartbeat must run between the PLC and the TwinSAFE controller.

- If the heartbeat is lost (PLC hang, crash, or communication dropout), TwinSAFE autonomously triggers an emergency shutdown without requiring a PLC command.
- This ensures the system never depends on a healthy PLC to reach a safe state, and is the safety backstop for the PLC-request architecture in Section 5.

- Recovery from a watchdog-triggered emergency follows the same physical recovery as any other emergency (Section 6), even after the PLC has recovered.
- Watchdog timeout value: [PLACEHOLDER — to be agreed between the Mentee PLC engineer and Total Control].

8. PLC Signal Exposure

Currently only button status is exposed to the PLC. The following functions must be exposed so they can be requested by the Mentee software, each executed by the TwinSAFE controller. Per the governing principle in Section 5, UP-step requests are accepted but cover-gated by TwinSAFE; opening the cover and clearing an emergency are excluded. TwinSAFE also reports cover/latch status to the PLC. Each signal must be individually addressable; proposed names are for clarity — Total Control should align with their existing naming convention.

#	Signal (proposed)	Physical Equivalent	Behaviour / Gating
1	**CMD_RUN_TO_R&D**	Short press SAFETY RESET (in Run)	Run → R&D. Sheds HV. Executed only after zero-speed + zero-current confirmed.
2	**CMD_R&D_TO_SAFE**	Short press SAFETY RESET (in R&D)	R&D → Safe. Sheds LV (de-energises LV domain).
3	**CMD_SAFE_TO_R&D**	Dual press (in Safe)	Safe → R&D (UP). Request only; TwinSAFE executes solely if cover CLOSED + LOCKED, else rejected.
4	**CMD_R&D_TO_RUN**	Dual press (in R&D)	R&D → Run (UP). Request only; TwinSAFE executes solely if cover CLOSED + LOCKED, else rejected.
5	**CMD_TRIGGER_EMERGENCY**	EMERGENCY STOP	Any → Emergency. Executed by TwinSAFE. LATCHING — not reversible via PLC; physical recovery required (Section 6). Supplementary to the hardwired E-Stop, which is unchanged.
6	**STS_COVER_LOCKED (status)**	— (read only)	TwinSAFE → PLC status report of the cover/latch (closed + locked). Read-only; lets the software check the UP gate before requesting.

Note: UP-step requests (signals 3 and 4) are accepted from the PLC but executed by TwinSAFE only when the cover is closed + locked — a request with the cover open is silently rejected, exactly as a button press would be. There is intentionally no PLC signal to open the cover or to clear an emergency; both remain physical-only.

9. Implementation Notes

- All software-initiated transitions are PLC requests executed by TwinSAFE; the PLC cannot bypass the safety controller.
- The hardwired physical E-Stop safety function is unchanged. The PLC emergency request is supplementary and must not re-route the E-Stop through the PLC.
- Any transition that drops HV (leaving Run Mode) is gated on confirmed zero motor speed and zero current.
- No additional hardware changes are anticipated; changes are within the TwinSAFE / PLC software and configuration. Total Control to confirm during review.
- Electrical drawings must be updated to reflect the revised state structure, signal exposure, and watchdog requirement.
- Items for Total Control / Mentee PLC engineer to confirm during review: watchdog timeout value (Section 7); dual-press timing (simultaneous hold duration to avoid accidental activation); and that R&D Mode re-enabling the full LV domain is consistent with the existing emergency drop sequence.

10. Document Change History

This section records the evolution of the requirement from Total Control's original logic through the current revision, so reviewers can see what changed and why.

Ver.	Basis	Change
—	Original	Total Control as-built logic: three buttons, single operating condition plus Emergency. Short press SAFETY RESET → dead safe state; long press → re-activate all power supplies; RELEASE LOCK opens cover with green flashing.
1.0	Mentee + Mark	Introduced Prep / Run terminology and free cycling between them. Added staged power-up (LV then HV) and software PIN concept (later dropped). PLC command exposure and PLC–TwinSAFE watchdog/heartbeat introduced.
1.1	Mentee	Refined gesture model to dual-press = UP / short press = DOWN. Added de-energised Safe Mode and Post-E-Stop state. Governing safety principle introduced: PLC may shed energy but not add energy or open cover. Renamed Prep → R&D Mode.
1.2	Mentee	Set UP-steps and cover-open to physical-only. Reinstated PLC signal table limited to safer-direction transitions plus the no-physical-change permission step. Reworded recovery so physical-only applies to energising/access steps.
1.3	Mentee	Removed the Post-E-Stop state as redundant: dis-engaging the E-Stop now returns directly to Safe Mode. Recovery collapsed to a direct Emergency → Safe transition; corresponding PLC signal removed. State count reduced from five to four plus Emergency.
1.4	Mentee	Restored PLC ability to request UP-steps (Safe → R&D, R&D → Run), gated by TwinSAFE on cover closed + locked and rejected if open — identical to button behaviour. Added cover/latch status reporting (STS_COVER_LOCKED) to the PLC. Revised governing principle to the request/filter model. All UP-steps (physical or PLC) now require the cover closed.